B.I.R.D.S — Broad Igneous Rocks · Diagrams & Statistics

Application workflow

Geochemical dataset
.csv · .tsv · .xls · .xlsx or a saved session (.birds)

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Import
delimiter and encoding detected · decimal comma → dot · values below detection (<0.5, b.d.l.) converted

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Sample grouping
up to three grouping columns · names, colours and markers · groups shown or hidden

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Reference catalogue
15 reservoirs · 28 normalisers (REE chondrite 8, shale 6, multi-element 9, Ni-Cu-PGE 5)

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Analytical modules — one menu each

Classification (14)

TAS volcanic & plutonic · K_2O–SiO_2
AFM · Jensen · Hanski · Barker · Frost
A/CNK–A/NK · Ab–Or–Q
Th–Co · Zr/TiO_2–Nb/Y (altered rocks)

Bivariate

Individual and paired (Harker, matrix)
OLS regression with 95 % band
Reservoir overlays · group hulls

Spider diagrams

REE and multi-element patterns
28 normalisers, Ni-Cu-PGE included
Sample lines or group medians

Tectonic discrimination (34)

Pearce · Saccani · Pearce & Gale
Pearce et al. 1984 · Harris · Wood
Defant & Drummond · Yang (FCMS)
Vermeesch · Verma DF with probabilities

Statistics & anomalies (8 tabs)

Descriptive · by group · paper table
Eu/Eu*, Ce/Ce* … · trace ratios
Correlation matrix
208 tectonic parameters

Isotopes

Sr–Nd with εNd, εSr
and initial ratios
Pb–Pb (^{207}Pb, ^{208}Pb) with the NHRL

Petrography (5)

QAPF plutonic and volcanic
Gabbroic · ultramafic
Pyroclastic tuffs

CIPW norm

Classical CIPW (1903)
Normative minerals as new columns,
ready for any diagram

Map & kriging

Decimal · DMS · UTM · MGRS
Ordinary kriging, per group
Variogram · cross-validation
Grids for a GIS (.asc)

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Outputs
Plots: .png · .jpg · .pdf · .svg (72–1200 dpi) · statistics report: .xlsx · .csv
Kriged grids for a GIS: .asc + .prj · sessions: .birds

Implemented in Python 3 with PySide6, matplotlib, pandas, NumPy and SciPy. Nothing is downloaded once installed.

References: Rollinson & Pease (2021) · Pearce (2008) · Vermeesch (2006) · Verma & co-authors (2006–2017) · Yang et al. (2019) · Matheron (1963)

Input · Core layer · Analytical module · Output